

1_10xHuman_8CLC_Stellarscope

August 31, 2026

```
[1]: library(Seurat)
library(ggplot2)
library(ggpubr)
library(pheatmap)
library(dplyr)
library(RColorBrewer)
getwd()
dir.create("figures_10xHuman_8CLC")
dir.create("data_10xHuman_8CLC")

dataset_id <- "10xHuman_8CLC"
sample <- "DUX4"
```

Loading required package: SeuratObject

Loading required package: sp

‘SeuratObject’ was built under R 4.3.3 but the current version is 4.4.1; it is recommended that you reinstall ‘SeuratObject’ as the ABI for R may have changed

‘SeuratObject’ was built with package ‘Matrix’ 1.6.4 but the current version is 1.7.5; it is recommended that you reinstall ‘SeuratObject’ as the ABI for ‘Matrix’ may have changed

Attaching package: ‘SeuratObject’

The following objects are masked from ‘package:base’:

intersect, t

Warning message:

"package ‘ggplot2’ was built under R version 4.4.3"

Warning message:
"package 'dplyr' was built under R version 4.4.3"

Attaching package: 'dplyr'

The following objects are masked from 'package:stats':

filter, lag

The following objects are masked from 'package:base':

intersect, setdiff, setequal, union

Warning message:
"package 'RColorBrewer' was built under R version 4.4.3"
'/mnt/TEdata/TEbenchmarking/jupyterNotebooks/realDatasets'

Warning message in dir.create("figures_10xHuman_8CLC"):
"'figures_10xHuman_8CLC' already exists"
Warning message in dir.create("data_10xHuman_8CLC"):
"'data_10xHuman_8CLC' already exists"

```
[2]: path <- paste0("/mnt/TEresults2/snakemake_results/results/stellarscope_out/",  
  ↪ dataset_id, "/", sample, "/pseudobulk/")  
  
STAR_path <- paste0("/mnt/TEresults/snakemake_results/results/STARoutdir/  
  ↪ ", dataset_id, "/", sample, "/best_Solo.out/Gene")  
  
filteredBarcodes <- read.table(paste0(STAR_path, "/filtered/barcodes.tsv"))$V1  
  ↪ # read barcodes selected by STARsolo  
  
twoCLC_TEmatrix <- Seurat::ReadMtx(mtx = paste0(path, sample,  
  ↪ "_pseudobulk-TE_counts.mtx"),  
  cells = paste0(path, sample,  
  ↪ "_pseudobulk-barcodes.tsv"),  
  features = paste0(path, sample,  
  ↪ "_pseudobulk-features.tsv"),  
  feature.column = 1) # read matrix  
twoCLC_TEmatrix <-  
  ↪ twoCLC_TEmatrix[setdiff(rownames(twoCLC_TEmatrix), "__no_feature"), filteredBarcodes]
```

```

nCells <- ncol(twoCLC_TEmatrix)
thrMinCells <- round(nCells * 0.02)
thrMinCells

# create Seurat object with shallow filtering of TEs expressed in at least 20
  ↪ cells and cells expressing at least 50 TEs
objTE_stellarscope <- Seurat::CreateSeuratObject(twoCLC_TEmatrix, project =
  ↪ "8CLC",
                                                min.cells = thrMinCells, min.features = 0)

#objTE_stellarscope <- objTE_allCB[,filteredBarcodes] # keep only filtered
  ↪ barcodes

objTE_stellarscope

```

2

Warning message:

"Feature names cannot have underscores ('_'), replacing with dashes ('-')"

An object of class Seurat

234933 features across 125 samples within 1 assay

Active assay: RNA (234933 features, 0 variable features)

1 layer present: counts

[3]: `summary(rowSums(objTE_stellarscope@assays$RNA$counts > 0))`

```

      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
      2.00   2.00   3.00   4.93   5.00  114.00

```

[4]: `annotation_stellarscope <- read.table("/mnt/TEdata/TEbenchmarking/Rscripts/
 ↪ annotation/annotation_stellarscope_hg38.tsv")
head(annotation_stellarscope)`

A data.frame: 6 × 10

	chr	start	end	sw_score	strand	locusID	family	subfamily
	<chr>	<int>	<int>	<int>	<chr>	<chr>	<chr>	<chr>
1	chr1	67108754	67109046	1892	+	L1P5	L1	L1P5
2	chr1	8388316	8388618	2582	-	AluY	Alu	AluY
3	chr1	25165804	25166380	4085	+	L1MB5	L1	L1MB5
4	chr1	33554186	33554483	2285	-	AluSc	Alu	AluSc
5	chr1	41942895	41943205	2451	-	AluY_dup1	Alu	AluY
6	chr1	50331337	50332274	1587	+	HAL1	L1	HAL1

[5]: `# Remove some classes of TEs`

```

classesToExclude <- c("Other", "Satellite", "Unknown", "RNA")

```

```

stellarscopeTEs <- Features(objTE_stellarscope)

```

```
TEsToKeep <- annotation_stellarscope[! annotation_stellarscope$class %in%
  ↳ classesToExclude, ]$stellarscopeID

length(setdiff(stellarscopeTEs, TEsToKeep))/ length(stellarscopeTEs) * 100 #
  ↳ print percentage removed

objTE_stellarscope <- objTE_stellarscope[intersect(stellarscopeTEs, TEsToKeep),]
```

0.0770432421158373

```
[6]: feature_metadata <- annotation_stellarscope[match(Features(objTE_stellarscope),
  ↳ annotation_stellarscope$stellarscopeID),]

head(feature_metadata)
```

A data.frame: 6 × 10

	chr	start	end	sw_score	strand	locusID	family	subfa
	<chr>	<int>	<int>	<int>	<chr>	<chr>	<chr>	<chr>
2	chr1	8388316	8388618	2582	-	AluY	Alu	AluY
9	chr1	92274206	92275925	536	+	L2	L2	L2
20	chr1	1048489	1048789	2439	+	AluSp	Alu	AluS
66	chr1	114294733	114297115	4561	+	HAL1_dup1	L1	HAL
79	chr1	156237768	156238043	1887	-	AluSx1_dup2	Alu	AluS
87	chr1	174063389	174063739	355	+	L2_dup2	L2	L2

```
[7]: options(repr.plot.width=7, repr.plot.height=5)

objTE_stellarscope@meta.data$nCount_TE <- objTE_stellarscope@meta.
  ↳ data$nCount_RNA
objTE_stellarscope@meta.data$nFeature_TE <- objTE_stellarscope@meta.
  ↳ data$nFeature_RNA
# Visualize QC metrics as a violin plot
VlnPlot(objTE_stellarscope, features = c("nCount_TE"), ncol = 1,
  pt.size = 0) + theme(text=element_text(size=17))
ggsave(paste0("figures_", dataset_id, "/nCountTE_violin_stellarscope.pdf"),
  ↳ device='pdf')

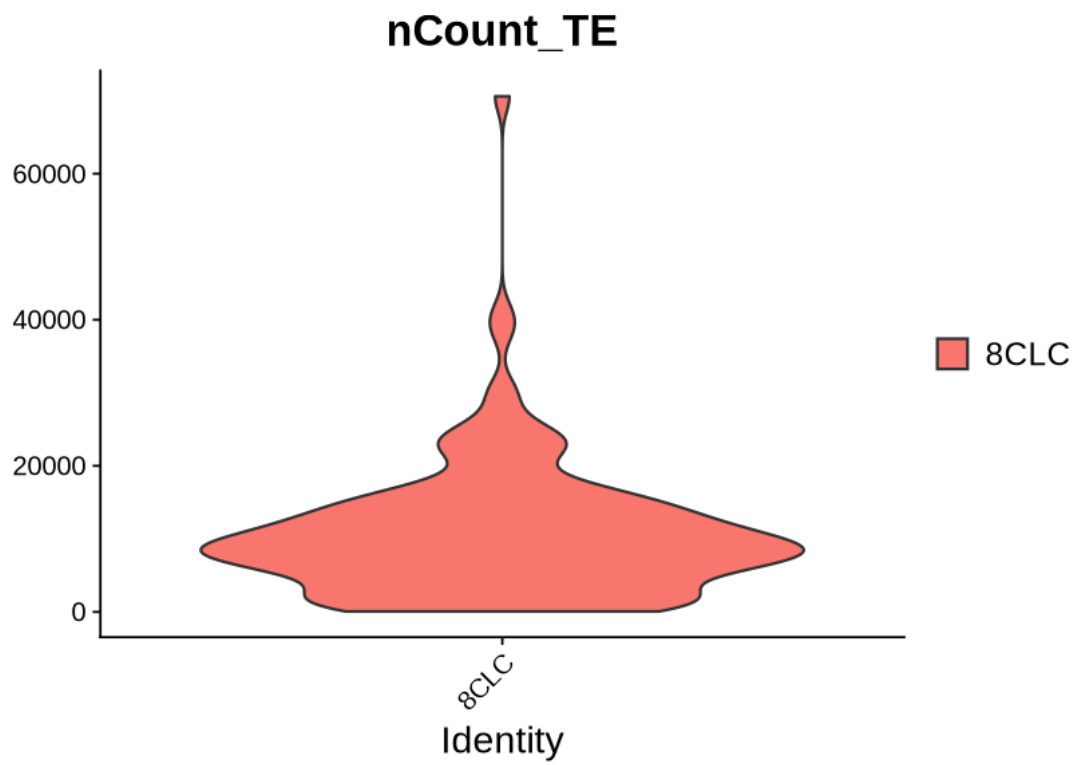
VlnPlot(objTE_stellarscope, features = c("nFeature_TE"), ncol = 1,
  pt.size = 0) + theme(text=element_text(size=17))
ggsave(paste0("figures_", dataset_id, "/nFeatureTE_violin_stellarscope.pdf"),
  ↳ device='pdf')
```

Warning message:

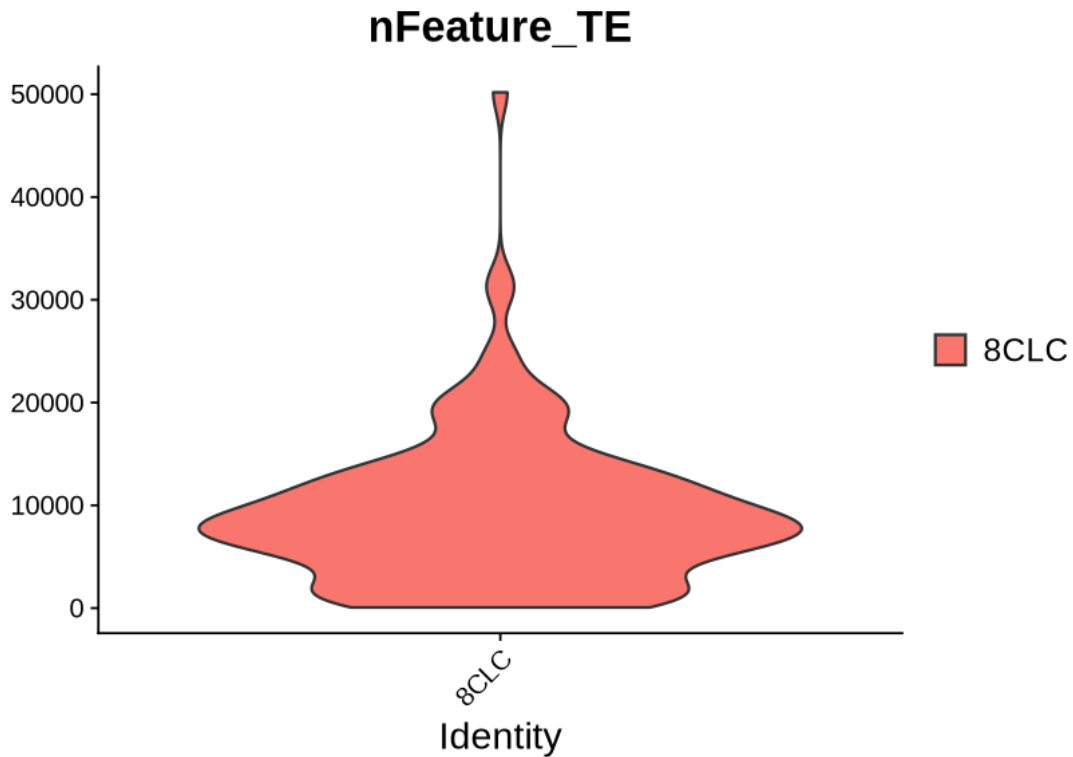
"Default search for "data" layer in "RNA" assay yielded no results; utilizing
"counts" layer instead."

Warning message:

```
"The `slot` argument of `FetchData()` is deprecated as of SeuratObject
5.0.0.
  Please use the `layer` argument instead.
  The deprecated feature was likely used in the Seurat
package.
  Please report the issue at
https://github.com/satijalab/seurat/issues."
Warning message:
"`PackageCheck()` was deprecated in SeuratObject 5.0.0.
  Please use `rlang::check_installed()` instead.
  The deprecated feature was likely used in the Seurat
package.
  Please report the issue at
https://github.com/satijalab/seurat/issues."
Warning message:
"`aes_string()` was deprecated in ggplot2 3.0.0.
  Please use tidy evaluation idioms with `aes()`.
  See also `vignette("ggplot2-in-packages")` for more information.
  The deprecated feature was likely used in the Seurat
package.
  Please report the issue at
https://github.com/satijalab/seurat/issues."
Saving 6.67 x 6.67 in image
Warning message:
"Default search for "data" layer in "RNA" assay yielded no results; utilizing
"counts" layer instead."
```



Saving 6.67 x 6.67 in image



```
[8]: objTE_stellarscope <- JoinLayers(objTE_stellarscope)

objTE_stellarscope <- NormalizeData(objTE_stellarscope, normalization.method = "LogNormalize",
  ↪ scale.factor = 10000)

objTE_stellarscope <- FindVariableFeatures(objTE_stellarscope, selection.method = "vst",
  ↪ nfeatures = 4000)

# Identify the 10 most highly variable genes
top10 <- head(VariableFeatures(objTE_stellarscope), 10)

# plot variable features with and without labels
plot1 <- VariableFeaturePlot(objTE_stellarscope)
plot2 <- LabelPoints(plot = plot1, points = top10, repel = TRUE)
plot2
```

Normalizing layer: counts

Finding variable features for layer counts

Warning message in simpleLoess(y, x, w, span, degree = degree, parametric = parametric, :

```

"at -1.8141"
Warning message in simpleLoess(y, x, w, span, degree = degree, parametric =
parametric, :
"radius 0.00033171"
Warning message in simpleLoess(y, x, w, span, degree = degree, parametric =
parametric, :
"all data on boundary of neighborhood. make span bigger"
Warning message in simpleLoess(y, x, w, span, degree = degree, parametric =
parametric, :
"pseudoinverse used at -1.8141"
Warning message in simpleLoess(y, x, w, span, degree = degree, parametric =
parametric, :
"neighborhood radius 0.018213"
Warning message in simpleLoess(y, x, w, span, degree = degree, parametric =
parametric, :
"reciprocal condition number 1"
Warning message in simpleLoess(y, x, w, span, degree = degree, parametric =
parametric, :
"There are other near singularities as well. 0.01561"
Warning message in simpleLoess(y, x, w, span, degree = degree, parametric =
parametric, :
"zero-width neighborhood. make span bigger"
Rasterizing points since number of points exceeds 100,000.
To disable this behavior set `raster=FALSE`

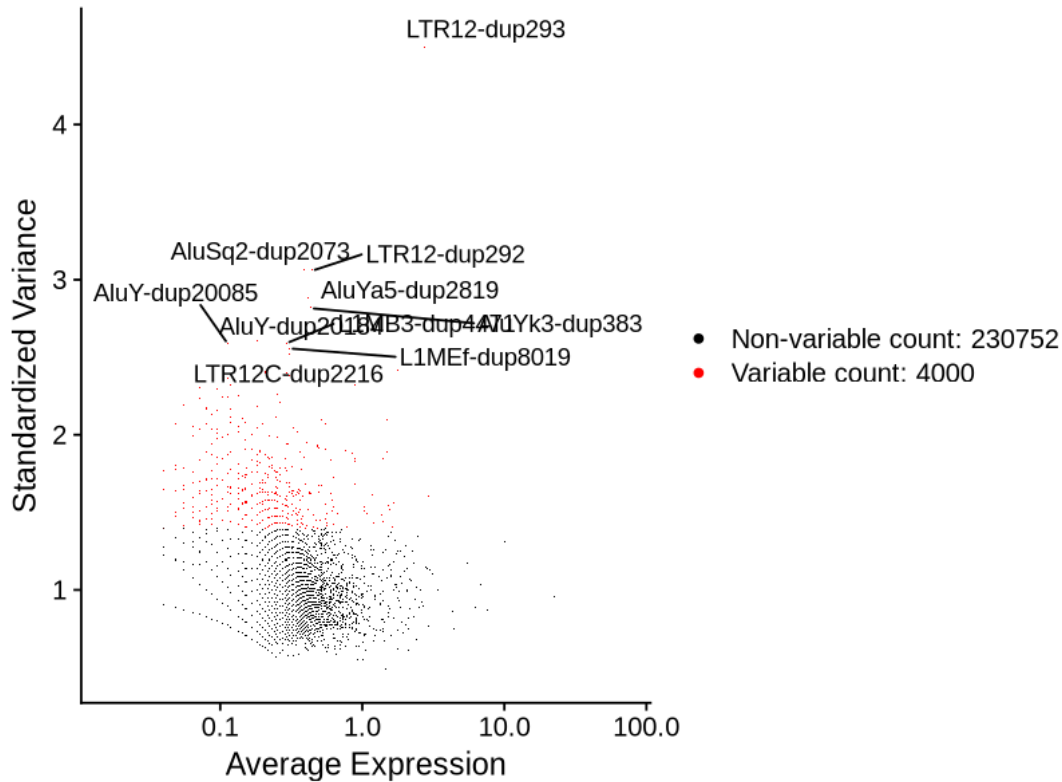
```

When using `repel`, set `xnudge` and `ynudge` to 0 for optimal results

```

Warning message:
"Removed 125543 rows containing missing values or values outside the
scale range
(`geom_scattermore()`)."

```

```
[9]: objTE_stellarscope
```

An object of class Seurat
 234752 features across 125 samples within 1 assay
 Active assay: RNA (234752 features, 4000 variable features)
 2 layers present: counts, data

```
[10]: gc()
all.genes <- rownames(objTE_stellarscope)
objTE_stellarscope <- ScaleData(objTE_stellarscope) # on hugs
```

		used	(Mb)	gc trigger	(Mb)	max used	(Mb)
A matrix: 2 × 6 of type dbl	Ncells	18475097	986.7	47609052	2542.6	36587912	1954.1
	Vcells	115090634	878.1	241071378	1839.3	240984085	1838.6

Centering and scaling data matrix

```
[11]: grep("ERVL", all.genes, value = T)[1:20]
```

1. 'ERVL-B4-int-dup5' 2. 'ERVL-E-int-dup75' 3. 'ERVL-E-int-dup76' 4. 'ERVL-E-int-dup77'
 5. 'ERVL-E-int-dup124' 6. 'ERVL-E-int-dup127' 7. 'ERVL-B4-int-dup47' 8. 'HERVL40-int-dup10'
 9. 'HERVL40-int-dup12' 10. 'HERVL40-int-dup13' 11. 'ERVL-E-int-dup472' 12. 'ERVL-E-in-

t-dup559' 13. 'HERVL-int-dup142' 14. 'ERVL-E-int-dup602' 15. 'ERVL-E-int-dup604'
 16. 'HERVL-int-dup166' 17. 'ERVL-E-int-dup691' 18. 'ERVL-E-int-dup706' 19. 'ERVL-E-int-
 t-dup748' 20. 'ERVL-E-int-dup753'

```
[12]: objTE_stellarscope <- RunPCA(objTE_stellarscope, features =  

  ↪VariableFeatures(object = objTE_stellarscope))  
  

  DimPlot(objTE_stellarscope, reduction = "pca") + NoLegend()  
  

  ElbowPlot(objTE_stellarscope)
```

PC_ 1

Positive: AluSp-dup33484, AluSz-dup27466, L1ME4b-dup5582, AluSp-dup43257,
 AluJb-dup115959, AluJb-dup3035, AluY-dup104921, MER57-int-dup1085,
 AluSg7-dup7208, AluSx1-dup109094

L1PA14-dup1753, AluSx1-dup32080, AluSx1-dup59414, L2b-dup4964,
 AluYj4-dup3393, AluSq2-dup58595, L1PBa-dup1608, AluJb-dup43522, AluSq-dup11127,
 L1MD1-dup4962

AluSz-dup64611, AluSz-dup8044, L1MB4-dup2112, HAL1-dup7399,
 L1MC4-dup21123, AluSz-dup96759, AluSx-dup38015, MER57-int-dup1089,
 AluSc5-dup512, MER3-dup8939

Negative: L1MEc-dup10283, L1MB4-dup4324, L1PA3-dup2755, L1PA6-dup3355,
 MIR3-dup8006, L1PA6-dup3347, AluSx-dup606, AluSx1-dup106460, AluY-dup51630,
 AluSg-dup24177

AluSx1-dup16560, L1MC2-dup4950, AluSc8-dup10299, AluSx1-dup47150,
 L1PA7-dup1108, AluSg-dup35799, L1MDa-dup2850, AluSz-dup49639, L1MC2-dup3115,
 AluSc8-dup15451

AluJr4-dup15102, MIR-dup145976, AluSx1-dup10463, L1MC1-dup1280,
 L1MC3-dup6714, MSTB-dup6217, L1PA10-dup717, L1MC1-dup6842, MIR-dup146442,
 AluSx1-dup24803

PC_ 2

Positive: L1M4c-dup1017, L1PA16-dup3175, L1PA10-dup1802, LTR12-dup293, L1MCa-
 dup2430, MIR3-dup8006, L1MEf-dup9631, AluJb-dup35468, MLT1D-dup5155,
 AluSx1-dup109096

HERV17-int-dup775, LTR12-dup292, AluSx1-dup109115, AluJr-dup28737,
 L1PB1-dup13537, L1PA13-dup4187, MER5B-dup9535, L1PA5-dup9249, AluSx1-dup109094,
 L1PA5-dup3855

L2c-dup111605, L1M5-dup35479, AluSq2-dup14961, L1PA16-dup10946,
 MER68-int-dup316, L1MA6-dup2905, L1PBa-dup1092, L1PB4-dup5359, L1PA7-dup4466,
 L1P2-dup1198

Negative: AluSq2-dup42737, MIRc-dup57896, THE1C-dup3458, L1MEc-dup9914,
 L1PA16-dup11137, L1PREC2-dup1372, AluSz6-dup29625, L2a-dup115776, AluY-dup20085,
 Tigger20a-dup1022

HAL1-dup1, AluSz6-dup38539, MLT2A1-dup751, AluSx1-dup75950,
 MLT2A1-dup2942, L1MA10-dup3240, Tigger3a-dup3712, SVA-D-dup926, THE1D-dup10305,
 HERVL-int-dup1789

L2a-dup160055, AluSp-dup4850, L1ME4b-dup573, AluSz6-dup30317,
 L2a-dup129456, AluSg-dup5397, L1PA13-dup2972, AluJo-dup3737, AluSx1-dup100436,

MLT2A2-dup495

PC_ 3

Positive: AluJr4-dup7027, L1MA4A-dup2209, L1ME2-dup2239, L1MB7-dup23883, L1PA10-dup717, L1MC4-dup21935, AluSp-dup8973, AluSx4-dup172, AluSx1-dup77298, L1PA10-dup2558

L1PA4-dup11637, AluSx3-dup27520, AluSx4-dup892, L1MA4A-dup74, AluY-dup7965, L1ME2z-dup2302, AluSq2-dup10705, AluSx-dup8702, L1M3-dup1523, AluSx-dup101986

AluSg7-dup5696, AluSx-dup93736, Tigger3b-dup536, L1PA3-dup1801, AluSq2-dup5594, AluSx-dup34064, AluJr-dup45299, AluSq2-dup10619, AluYm1-dup1490, LTR12C-dup1085

Negative: AluSq2-dup51139, MER58B-dup3909, AluSx-dup598, AluJo-dup5591, AluSx1-dup14889, AluJr-dup70716, AluJb-dup16141, Charlie2a-dup1894, L1MA9-dup820, MLT2A2-dup3131

L1MB7-dup18204, AluSg-dup19930, L1PB1-dup272, L1MA7-dup5251, AluSq-dup4978, L1MB8-dup1317, AluSq2-dup51321, L1MA1-dup154, L2a-dup129456, AluSp-dup26854

AluSx1-dup71018, AluY-dup4088, L1ME2z-dup3952, L1MB3-dup4471, L1PA13-dup2972, AluY-dup102283, AluJb-dup65248, AluSx3-dup19670, L1PREC2-dup1373, AluSx-dup28816

PC_ 4

Positive: AluSx-dup89908, AluSx-dup81782, L1PA7-dup1721, AluJb-dup3035, AluY-dup104921, AluSx1-dup8786, AluSg-dup2874, AluY-dup639, AluSx6-dup31190, AluSx1-dup77298

AluSp-dup3787, AluSx-dup8810, AluSc8-dup3418, LTR7Y-dup213, L1PA8-dup4109, AluSx-dup75220, L1MA4A-dup3008, L1MA4A-dup2209, THE1B-dup19689, AluSg7-dup9079

AluSp-dup17040, AluY-dup79799, AluSx6-dup22364, AluYf1-dup167, AluSx1-dup115415, AluSg-dup16144, L1MD1-dup375, AluSx1-dup90899, L1M5-dup46229, MLT1A0-dup21652

Negative: AluSq-dup2160, AluSq-dup16093, Tigger1-dup5505, L1PA14-dup1753, L1M4b-dup100, L2b-dup4964, L1M4-dup2607, AluSx-dup3542, AluJb-dup12314, AluSx-dup68560

L2-dup672, AluSq2-dup58595, AluSq-dup11127, AluSx1-dup19535, Charlie1-dup1297, AluSg7-dup7208, MER3-dup8939, AluSx-dup70498, FLAM-A-dup10350, MIR-dup146870

MER5C1-dup23, L1ME4b-dup5582, AluYm1-dup924, L1PBa-dup1045, L1MD1-dup4962, L1MA4-dup5420, AluSx-dup89589, L2a-dup155929, L1M5-dup2496, AluSg-dup6320

PC_ 5

Positive: AluJb-dup3035, AluY-dup104921, L2c-dup23121, L1PBa-dup1608, AluJb-dup43522, AluSp-dup43257, AluSp-dup46122, AluSx3-dup2523, AluSx1-dup115415, AluSx1-dup36988

AluSx4-dup172, L1M4-dup15873, AluSx-dup6144, AluSg-dup2874, AluSq2-dup44563, L1PA7-dup1721, AluSq-dup3819, L1ME3E-dup4345, AluJo-dup9542, AluSx-dup89908

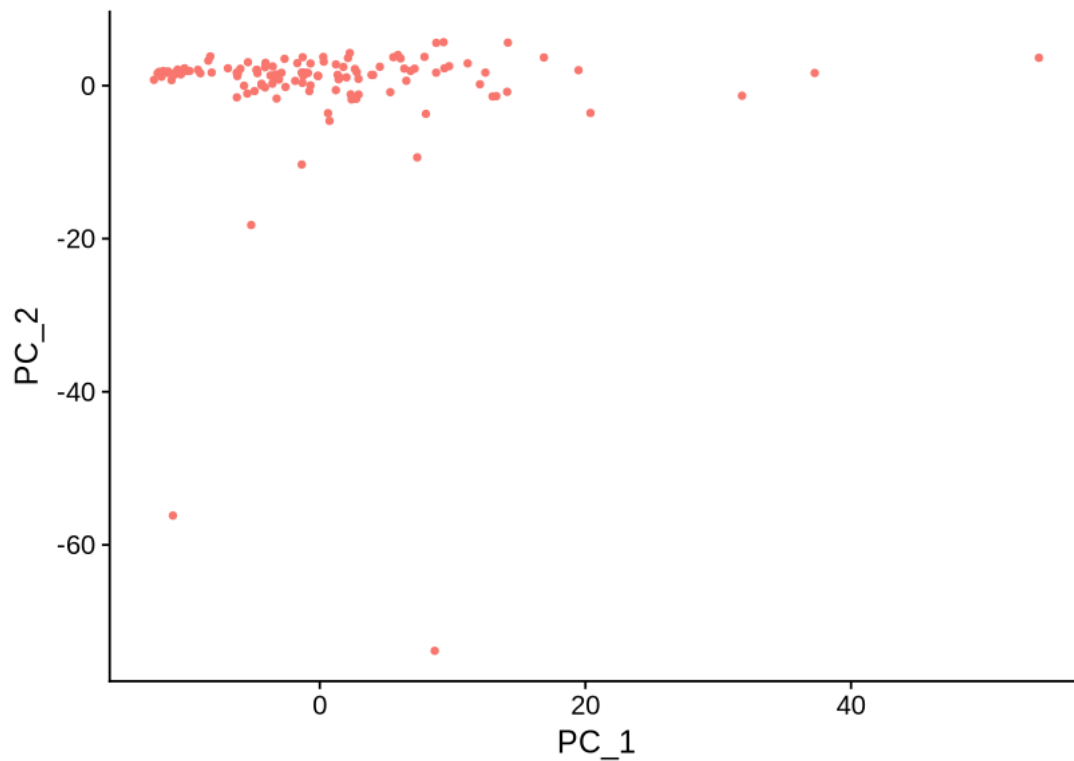
AluSq2-dup13265, AluSx6-dup22364, AluY-dup79799, AluSp-dup15526, L1PA6-dup1511, FLAM-A-dup534, AluY-dup71362, AluYh3a3-dup83, AluSx-dup83520,

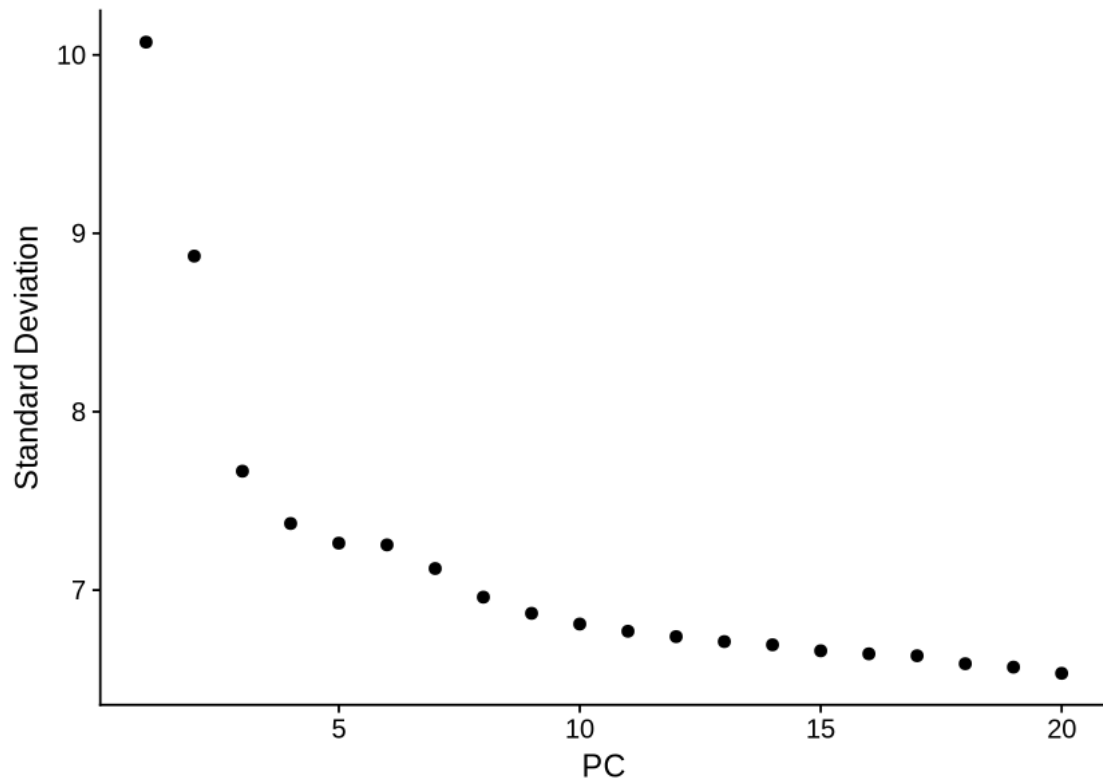
AluSx-dup101986

Negative: AluSz-dup34064, L1PA3-dup1801, L1ME2z-dup2302, L1MA4A-dup74,
AluSq2-dup51850, AluJr-dup45299, AluSx-dup8702, AluSz-dup9833, Tigger3b-dup536,
AluSq2-dup10619

AluSg-dup5977, AluSq2-dup5594, L1ME4a-dup1470, AluY-dup478,
MIR3-dup25208, AluSx3-dup23343, AluSc-dup28085, AluSx1-dup93259, L1M2-dup8768,
AluSx-dup17989

AluSc-dup28418, L1MB7-dup6187, AluSq2-dup34686, L1MD2-dup8969, MIRb-
dup217547, AluY-dup33876, MIRb-dup179850, AluJb-dup10460, AluSc-dup4614, AluSg-
dup16582





```
[13]: objTE_stellarscope <- FindNeighbors(objTE_stellarscope, dims = 1:10, k.param = 20)
      objTE_stellarscope <- FindClusters(objTE_stellarscope, resolution = 1, algorithm=4)
      objTE_stellarscope <- RunUMAP(objTE_stellarscope, dims = 1:10)
      DimPlot(objTE_stellarscope, reduction = "umap")
```

Computing nearest neighbor graph

Computing SNN

```
Warning message in RunLeiden(object = object, method = method, partition.type =
"RBConfigurationVertexPartition", :
"`random.seed` must be greater than 0 for leiden clustering, resetting
`random.seed` to 1."
Warning message:
"The default method for RunUMAP has changed from calling Python UMAP via
reticulate to the R-native UWOT using the cosine metric
To use Python UMAP via reticulate, set umap.method to 'umap-learn' and metric to
'correlation'
This message will be shown once per session"
```


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15:49:01 Writing NN index file to temp file
/mnt/TEdataStorage_2T/tmp/RtmpRrU1lr/file3ae0fe6197592

15:49:01 Searching Annoy index using 1 thread, search_k = 3000

15:49:01 Annoy recall = 100%

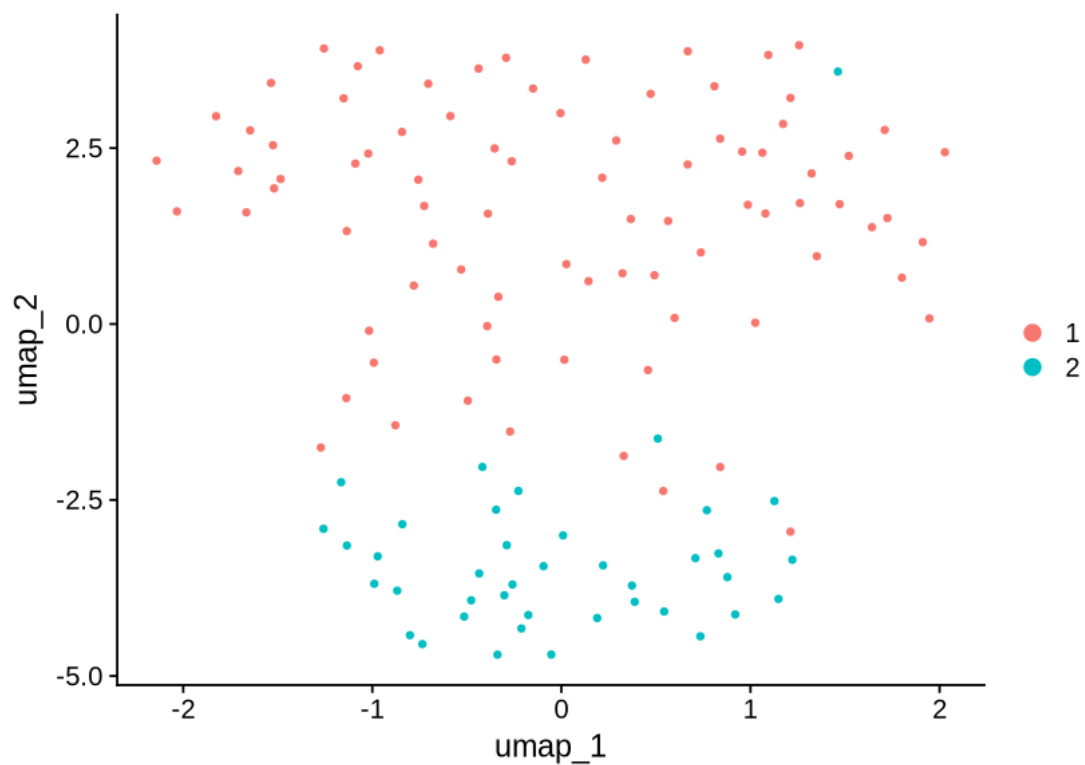
15:49:04 Commencing smooth kNN distance calibration using 1 thread
with target n_neighbors = 30

15:49:08 Initializing from normalized Laplacian + noise (using RSpectra)

15:49:08 Commencing optimization for 500 epochs, with 4568 positive edges

15:49:08 Using rng type: pcg

15:49:11 Optimization finished



```
[14]: objTE_stellarscope
```

```
An object of class Seurat
234752 features across 125 samples within 1 assay
Active assay: RNA (234752 features, 4000 variable features)
3 layers present: counts, data, scale.data
2 dimensional reductions calculated: pca, umap
```

```
[15]: FeaturePlot(objTE_stellarscope, reduction = "umap", features = "nCount_TE", pt.
      ↪size = 0.5) +
      theme_void() +
      theme(text=element_text(size=20))
```


nCount_TE



```
[16]: saveRDS(objTE_stellarscope, paste0("data_", dataset_id, "/stellarscope_",  
↪dataset_id, "_seuratObj.RDS"))
```